Figure 6.14 An example of a node where edges to children are stored in a dictionary. Dictionary's keys are characters, the first letter of each label, while values contain the full labels and the references to the destination node of the edge. The figure shows a simplification of the same searches in figure 6.1; as before, comparisons are based on the first character of the searched strings.

This solution is illustrated in figure 6.13, where we also show how binary search works to find the possible edge matching.

Notice that the match between the string searched and an edge's label doesn't have (and usually isn't) full; we'll see in a moment what this means for our algorithms and how to handle these situations.

Of course, using sorted arrays, as we have discussed in chapter 4 and appendix C, means logarithmic search, but linear (translated: slow!) insertion: although the number of elements can be considered a constant, in asymptotic analysis, from a practical point of view this implementation can significantly slow down insertion of new keys in large tries.

